

Simplified Analysis of the Auction Algorithm for Weighed Matroid Intersection By GPT 6 Astra

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Author's note

The proof analyzing our algorithm in the current arXiv version [2] was generated by GPT 5.6 Sol. We have carefully verified the proof and rewrote the manuscript [2] ourselves. However, we do not yet have a satisfactory intuitive understanding of the analysis. This manuscript, generated by GPT 6 Astra, provides a simpler and easier-to-follow analysis of the same algorithm. We intend to rewrite this simpler analysis ourselves, in our own words, and incorporate the resulting exposition into a future version of the manuscript on arXiv.

This draft was generated entirely by GPT 6 Astra.

Abstract

We simplify the analysis of the weighted matroid intersection auction algorithm using two familiar ideas: optimality under single-element basis exchanges, and the ordinary primal-dual gap. The exchange argument replaces the previous comparison of threshold spans. The gap contracts geometrically, up to a charge whose total is bounded by the accumulated owner price. A single logarithmic summation proves the phase bound $O(\varepsilon^{-1} \log(2n) + \varepsilon^{-2})$, without dividing the execution into two stages. The phase updates of the algorithm are unchanged.

1 Upper bounds, lower bounds, and the main statement

This revision develops the simpler analysis of [2], following the use of ordinary optimal-basis values and owner prices in Huang and Kobayashi [1, Section 5.3]. At a phase boundary, let

$$P = \pi(X), \quad Q = \max_{B \text{ basis of } \mathcal{M}_1} q(B), \quad D = P + Q.$$

We will show that $D \geq \text{OPT}$. If M is the weight of the best phase-boundary solution seen so far, then $M \leq \text{OPT}$. Thus

$$g = D - M \geq 0$$

is simply the gap between an upper bound and a lower bound on the optimum. There is an immediate approximation certificate:

$$g < 5\varepsilon D \implies M > (1 - 5\varepsilon)D \geq (1 - 5\varepsilon)\text{OPT}. \quad (1)$$

The key estimate says that one phase moves the upper bound toward the slightly inflated solution value $(1 + 2\varepsilon)w(X)$, apart from a bounded charge for price increases:

$$D' \leq \frac{D + \varepsilon(1 + 2\varepsilon)w(X)}{1 + \varepsilon} + 3(P' - P). \quad (2)$$

As long as the certificate fails, this implies

$$g' - g \leq -\frac{\varepsilon}{2(1+\varepsilon)}g + 3(P' - P). \quad (3)$$

The gap therefore contracts by a fixed fraction, except for price payments. Their total is bounded by $P \leq (1+\varepsilon)\text{OPT}$. This is the entire convergence mechanism.

Theorem 1. *Let $0 < \varepsilon \leq 1/10$. After removing loops in either matroid and nonpositive-weight elements, suppose that the ground set is nonempty. The original auction phase updates produce a boundary with $g < 5\varepsilon D$ within*

$$T_* = \left\lceil 3\varepsilon^{-1} \log(2n) + 2\varepsilon^{-2} \right\rceil = O(\varepsilon^{-1} \log(2n) + \varepsilon^{-2})$$

phases. Returning the best boundary state gives a $(1 - 5\varepsilon)$ -approximation. With this stopping budget and the original per-phase implementation, the number of independence-oracle queries is

$$O(n\varepsilon^{-1} \log^2(2n) + n\varepsilon^{-2} \log(2n)).$$

The original budget $\lceil 10\varepsilon^{-2} \log(2n) \rceil$ also suffices.

For a target ratio $1 - \eta$, use $\varepsilon = \eta/5$ for $0 < \eta \leq 1/2$, and $\varepsilon = 1/10$ for $1/2 < \eta < 1$. The theorem concerns scan phases. It does not require the algorithm to evaluate the certificate at its final boundary: the original rule of returning the best boundary state already suffices.

2 The algorithm and the properties we retain

2.1 Prices, utilities, and one phase

Let $\mathcal{M}_i = (E, \mathcal{I}_i)$, for $i \in \{1, 2\}$, be matroids on a common ground set, let $n = |E|$, and let $w : E \rightarrow \mathbb{R}_{>0}$. For a function f on E , write $f(S) = \sum_{e \in S} f(e)$; also write $S + e = S \cup \{e\}$ and $S - e = S \setminus \{e\}$. Let $\text{OPT} = \max\{w(S) : S \in \mathcal{I}_1 \cap \mathcal{I}_2\}$. Elements that are loops in either matroid, and nonpositive-weight elements if present, can first be discarded. The empty instance is immediate. We henceforth assume $n \geq 1$, so $w_{\max} := \max_{e \in E} w(e) \leq \text{OPT}$. All logarithms are natural.

If $X \in \mathcal{I}_i$, $e \notin X$, and $X + e \notin \mathcal{I}_i$, let $C_i(X, e)$ be the fundamental circuit in $X + e$. Set $\beta = 1 + \varepsilon$, $W = 2w_{\max}$, and

$$\Lambda = \{W\beta^{-k} : k = 0, 1, 2, \dots\}.$$

For $0 < u \leq w_{\max}$, let $\lceil u \rceil_{\Lambda}$ be the smallest member of Λ at least u , and set $\lceil 0 \rceil_{\Lambda} = 0$. Thus $u \leq \lceil u \rceil_{\Lambda} < \beta u$ for $u > 0$.

The algorithm maintains a common independent set X and owner labels $\pi(x), \kappa(x)$ for $x \in X$. Its effective prices and utilities are

$$p_X(e) = \begin{cases} \pi(e), & e \in X, \\ 0, & e \notin X, X + e \in \mathcal{I}_2, \\ \min\{\pi(f) : f \in C_2(X, e) - e\}, & e \notin X, X + e \notin \mathcal{I}_2, \end{cases}$$

and

$$q_X(e) = \begin{cases} \kappa(e), & e \in X, \\ \lceil (w(e) - p_X(e))_+ \rceil_{\Lambda}, & e \notin X, \end{cases} \quad (a)_+ = \max\{a, 0\}.$$

Owner labels remain unchanged as long as the element remains in X . In particular, owner utilities need not lie on the geometric grid.

Initially $X = \emptyset$. A phase consists of the following operations.

1. Freeze $X^* = X$ and $q^* = q_{X^*}$, set $Y = X^*$, and find a q^* -maximum basis B of $\mathcal{M}_1$ containing X^* .
2. Scan each element $e \in E \setminus X^*$ once, in any order. Put $\lambda = q^*(e)$, $p_e = p_X(e)$, and $u_e = w(e) - p_e$. Accept e if

$$B_{>\lambda} \cup Y_{=\lambda} \cup \{e\} \in \mathcal{I}_1, \quad u_e > 0, \quad u_e + \varepsilon w(e) \geq \lambda.$$

Here and below, threshold subscripts without an explicit weight function refer to q^* .

3. Upon accepting e , add it to Y . If $X + e \in \mathcal{I}_2$, add it to X ; otherwise replace a minimum- π element of $C_2(X, e) - e$ by e . Set

$$\pi(e) = p_e + \varepsilon w(e), \quad \kappa(e) = u_e + \varepsilon w(e).$$

The third step is performed immediately after each acceptance in the scan. Elements removed from X remain in Y . After the prescribed number of phases, return the heaviest phase-boundary set, including the initial empty set. These are exactly the phase updates of the original algorithm.

2.2 Elementary phase properties

For a fixed phase, let X', q' be its final state and utility function, and let Y now denote its final history set. Put $\mathcal{A} = Y \setminus X^*$. Thus $\mathcal{A}$ contains every element accepted in the phase, including elements that are subsequently removed from X .

We retain the following elementary properties from the original analysis [2]. They are established before its residual-value and potential arguments.

- (i) At every point of the phase, $X \subseteq Y$, $X \in \mathcal{I}_1 \cap \mathcal{I}_2$, and Y is contained in a q^* -maximum basis of $\mathcal{M}_1$. At a phase boundary, X is contained in a q_X -maximum basis.
- (ii) Each effective price $p_X(e)$ is nondecreasing throughout the execution. Accepting e increases $P = \pi(X)$ by exactly $\varepsilon w(e)$. Hence

$$P' - P = \varepsilon w(\mathcal{A}). \tag{4}$$

- (iii) For every current owner x ,

$$\pi(x) + \kappa(x) = (1 + 2\varepsilon)w(x), \quad 0 \leq \pi(x), \kappa(x) \leq \beta w(x).$$

Consequently $P, q_X(X) \leq \beta w(X) \leq \beta \text{OPT}$.

- (iv) The endpoint utilities satisfy

$$\begin{aligned} q'(e) &\leq q^*(e), & e \in X^*, \\ q'(e) &\leq q^*(e) + \varepsilon w(e), & e \in \mathcal{A}, \\ q'(e) &\leq q^*(e), & e \notin Y. \end{aligned}$$

- (v) If $e \notin Y$ and $Y + e$ is contained in a q^* -maximum basis, then

$$q'(e) \leq q^*(e)/\beta. \tag{5}$$

In words: an unaccepted element that remains compatible with the final history loses at least one utility level.

The utility upper bound in (iii) also follows directly from $\kappa(x) = w(x) - p_x + \varepsilon w(x)$ at acceptance and $p_x \geq 0$. For clarity, Appendix A recalls why the endpoint and compatibility statement hold, including when owners are removed or have utilities outside the grid.

Two details will be essential: positive $q^*(e)$ lies in Λ for every $e \notin X^*$, and $q^*(e) < \beta w(e)$ for every such element. Old owners, whose utilities may be outside Λ , all belong to Y .

3 The price bound from exchange optimality

A basis is maximum-weight if and only if no feasible single-element exchange increases its weight. For nonnegative weights, the corresponding criterion for an independent set is that neither an insertion nor a single-element exchange increases its weight. We use this standard matroid optimality criterion twice.

First, X is a maximum-price independent set in $\mathcal{M}_2$. An element that can be inserted into X has effective price zero. For a nonowner e that cannot be inserted, every exchangeable $x \in X$ belongs to $C_2(X, e) - e$, so $p_X(e) \leq \pi(x) = p_X(x)$. Consequently

$$p_X(I) \leq p_X(X) = P \quad (I \in \mathcal{I}_2).$$

Also $w(e) \leq p_X(e) + q_X(e)$ for every element: this follows from rounding upward for nonowners and from the owner-label identity for owners. For an optimal common independent set O ,

$$\text{OPT} = w(O) \leq p_X(O) + q_X(O) \leq P + Q = D. \quad (6)$$

Here $q_X(O) \leq Q$ because $q_X \geq 0$ and O extends to a basis of $\mathcal{M}_1$. This proves the upper bound without an integral or a separate lemma about threshold spans.

4 One phase, proved by exchanges

Lemma 2. *For one phase, with the notation of Section 2 and $\beta = 1 + \varepsilon$,*

$$Q' \leq \frac{Q}{\beta} + \frac{\varepsilon}{\beta} q^*(Y) + \varepsilon w(\mathcal{A}). \quad (7)$$

Proof. Choose a q^* -maximum basis C containing Y , and assign it allowances

$$d(c) = \begin{cases} q^*(c)/\beta, & c \in C \setminus Y, \\ q^*(c), & c \in X^*, \\ q^*(c) + \varepsilon w(c), & c \in \mathcal{A}. \end{cases}$$

These labels are used only in the proof. For $c \in C \setminus Y$, property (v) applies because $Y + c \subseteq C$; for $c \in Y$, property (iv) applies. Thus $q'(c) \leq d(c)$ for every $c \in C$.

Raise the final utilities on C to these allowances and leave the utilities outside C at q' . We claim that C is a maximum-weight basis for the resulting weights. Consider any feasible exchange $C - c + e$, where $e \notin C$. Since C was q^* -maximum, $q^*(e) \leq q^*(c)$. Also $e \notin Y$, so $q'(e) \leq q^*(e)$. It remains to check $q'(e) \leq d(c)$.

- If $c \in Y$, then $q'(e) \leq q^*(e) \leq q^*(c) \leq d(c)$.
- If $c \notin Y$ and $q^*(e) < q^*(c)$, the geometric grid gives $q^*(e) \leq q^*(c)/\beta = d(c)$. The case $q^*(e) = 0$ is immediate.
- If $c \notin Y$ and $q^*(e) = q^*(c)$, then $C - c + e$ is also q^* -maximum and contains $Y + e$. Property (v) gives $q'(e) \leq q^*(e)/\beta = d(c)$.

Thus no exchange improves C . Its value is $\sum_{c \in C} d(c)$, and we only raised final utilities, so

$$Q' \leq \sum_{c \in C} d(c) = \frac{Q}{\beta} + \frac{\varepsilon}{\beta} q^*(Y) + \varepsilon w(\mathcal{A}).$$

All comparisons that use the grid involve elements outside Y ; old owners with utilities outside the grid cause no difficulty. $\square$

Now $Y = X^* \dot{\cup} \mathcal{A}$, and $q^*(e) < \beta w(e)$ for $e \in \mathcal{A}$. Using $P' - P = \varepsilon w(\mathcal{A})$ in (7),

$$Q' \leq \frac{Q}{\beta} + \frac{\varepsilon}{\beta} q^*(X^*) + 2(P' - P).$$

The two copies of the price increase pay for protecting accepted elements from the discount and for their owner-utility bonuses. Adding P' and using $P + q^*(X^*) = (1 + 2\varepsilon)w(X^*)$ gives

$$D' \leq \frac{D}{\beta} + \frac{\varepsilon}{\beta} (1 + 2\varepsilon)w(X^*) + 3(P' - P),$$

which is (2).

5 One summation of the ordinary duality gap

Let $M_t = \max_{0 \leq j \leq t} w(X_j)$, $D_t = P_t + Q_t$, and $g_t = D_t - M_t$. Here $t = 0$ denotes the initial empty state, so $P_0 = M_0 = 0$. The quantities M_t and P_t are nondecreasing, and $D_t \geq \text{OPT} \geq M_t$. Since $w(X_t) \leq M_t \leq M_{t+1}$, (2) implies

$$g_{t+1} - g_t \leq -\frac{\varepsilon}{\beta} (g_t - 2\varepsilon M_t) + 3(P_{t+1} - P_t). \quad (8)$$

If the certificate has not yet succeeded, then $g_t \geq 5\varepsilon D_t$. Consequently $2\varepsilon M_t \leq 2\varepsilon D_t \leq (2/5)g_t$, so in particular $g_t - 2\varepsilon M_t \geq g_t/2$. This proves (3).

Suppose for contradiction that the certificate fails at every boundary $t = 0, \dots, T$. Then $g_t \geq 5\varepsilon D_t \geq 5\varepsilon \text{OPT} > 0$. Apply $\log x \leq x - 1$ to (3) and sum:

$$\begin{aligned} \log \frac{g_T}{g_0} &\leq -\frac{\varepsilon T}{2\beta} + 3 \sum_{t=0}^{T-1} \frac{P_{t+1} - P_t}{g_t} \\ &\leq -\frac{\varepsilon T}{2\beta} + \frac{3P_T}{5\varepsilon \text{OPT}} \leq -\frac{\varepsilon T}{2\beta} + \frac{3\beta}{5\varepsilon}. \end{aligned} \quad (9)$$

Initially $g_0 = Q_0 \leq \beta n \text{OPT}$, while $g_T \geq 5\varepsilon \text{OPT}$. Therefore

$$\begin{aligned} T &\leq \frac{2\beta}{\varepsilon} \log \frac{\beta n}{5\varepsilon} + \frac{6\beta^2}{5\varepsilon^2} \\ &\leq \frac{2\beta}{\varepsilon} \log n + \frac{8\beta^2}{5\varepsilon^2} < \frac{3}{\varepsilon} \log(2n) + \frac{2}{\varepsilon^2}. \end{aligned}$$

The second inequality uses $\log x \leq x$; the last uses $\beta \leq 1.1$. This contradicts $T = T_*$ and proves Theorem 1. The original phase budget also suffices because $3\varepsilon \log(2n) + 2 < 10 \log(2n)$.

The logarithm in (9) only measures the cumulative multiplicative contraction of a familiar quantity, $D - M$. There is no optimum-dependent shift, residual-value function, or separate initialization stage. Beyond the retained phase properties, the basis comparison uses just the usual exchange criterion. If one wants to check the certificate explicitly at the final boundary, one additional maximum-basis computation may be needed; this does not change the asymptotic query bound.

A Endpoint utilities and unaccepted compatible elements

We recall the short verifications of properties (iv) and (v), using the original basis invariant and price monotonicity. This makes explicit the cases involving removed owners.

Initial owners. If $e \in X^*$ survives the phase, its utility is unchanged. If it is removed, it cannot be accepted again in that phase because the scan excludes X^* . Its final effective price is at least its former owner price $\pi(e)$. If its final surplus is positive, then

$$q'(e) < \beta(w(e) - p_{X'}(e)) \leq \beta(w(e) - \pi(e)) \leq \kappa(e) = q^*(e).$$

The last inequality follows from $\kappa(e) - \beta(w(e) - \pi(e)) = \varepsilon(w(e) + \pi(e)) \geq 0$. If the final surplus is nonpositive, $q'(e) = 0$.

Elements accepted during the phase. Let $e \in \mathcal{A}$, and let p_e, u_e be its values at acceptance. Price monotonicity gives $p_e \geq p_{X^*}(e)$ and hence $u_e \leq q^*(e)$. If e survives, then $q'(e) = u_e + \varepsilon w(e) \leq q^*(e) + \varepsilon w(e)$. If it is removed later, price monotonicity gives $p_{X'}(e) \geq p_e + \varepsilon w(e)$. A positive final surplus therefore implies

$$q'(e) < \beta(w(e) - p_{X'}(e)) \leq \beta(u_e - \varepsilon w(e)) \leq u_e \leq q^*(e),$$

where the penultimate inequality uses $u_e \leq w(e)$. If the final surplus is nonpositive, $q'(e) = 0$. An element outside Y is never an owner during the phase, so price monotonicity directly gives $q'(e) \leq q^*(e)$. This proves property (iv).

Unaccepted compatible elements. Suppose $e \notin Y$ and $Y + e$ is contained in a q^* -maximum basis. Write $\lambda = q^*(e)$. If $\lambda = 0$, property (iv) settles the claim. Otherwise, the standard characterization of maximum bases says that $B_{>\lambda} \cup Y_{=\lambda} \cup \{e\}$ is independent. At the time e was scanned, the history was a subset of its final value, so the independence test for accepting e succeeded.

Since e was not accepted, either $u_e \leq 0$, which forces $q'(e) = 0$, or $u_e > 0$ and $u_e + \varepsilon w(e) < \lambda$. In the latter case,

$$\beta u_e \leq u_e + \varepsilon w(e) < \lambda,$$

because $u_e \leq w(e)$. Price monotonicity makes the final surplus at most $u_e < \lambda/\beta$. Since $\lambda/\beta \in \Lambda$, rounding upward still gives $q'(e) \leq \lambda/\beta$, as required.

References

- [1] Chien-Chung Huang and Yusuke Kobayashi. *Auction-Based Algorithms for Matroid Intersection: Near-Linear Query Complexity and Constant-Pass Semi-Streaming*. arXiv:2609.38685, 2026. <https://arxiv.org/abs/2609.38685>.
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